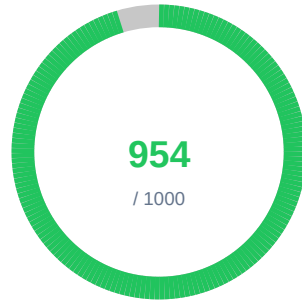


Code Hardener

ATTESTATION CERTIFICATE



bulletproof-conductor-kernel

Scan 1bc3abeb | Branch: main | Jul 24, 2026

Attestation ID

c50c22c5-5cbf-4900-b86d-...

Type

in-toto

Subject

bulletproof-conductor-kernel

Created

Jul 24, 2026, 4:21 PM

Subject Digest

sha256:ef060b56e7142551f9c96a4fe28da1dec944c1f656c7404900b822ce...

Signature Status

CRYPTOGRAPHICALLY SIGNED

Algorithm: Ed25519 (Local Signing Key)

Signature

4BtqhUJLAndZBy0igXoBqsJ8bt5XkEam0toDYI8qHXzKZ+GXfip28meuib2z4URT
ctnew2J91CBzmCYp+himBg==

This attestation certifies the security scan was performed and results were recorded with cryptographic integrity.

Scan Report — bulletproof-conductor-kernel

Branch: main | Status: completed | Jul 24, 2026

954 / 1000 Score — Quality: excellent

Scan Details

Profile: standard
Duration: 18s
Started: Jul 24, 2026, 4:21 PM
Completed: Jul 24, 2026, 4:21 PM
Files Analyzed: 148
Scanners Run: 11 of 12
Open Findings: 20
Resolved / Suppressed:0
Total Detected: 20
Codebase: Markdown: 70 files | Config: 43 files | Shell: 24 files | Python: 2 files | JavaScript/TypeScript: 1 files

0 Critical 0 High 4 Medium 2 Low 14 Info

Scanner Results

12 tools executed — 6 passed | 5 with open findings | 1 not applicable | 0 errors

Scanner	Category	Outcome	Findings	Duration	Files Scanned
trivy	SCA	FAIL	1	3s	1
gitleaks	Secrets	PASS	0	0s	148
opengrep	SAST	FAIL	2	6s	27
checkov	IaC	PASS	0	3s	43
grype	SCA	PASS	0	3s	148
syft	SBOM	FAIL	1	1s	1

package-validator	AI Code Quality	N/A	0	0s	148
oxlint	Code Quality	PASS	0	0s	1
ruff	Code Quality	FAIL	2	0s	2
actionlint	CI/CD Security	PASS	0	0s	43
jscpd	Code Quality	PASS	0	0s	3
typos	Code Quality	FAIL	14	0s	148

Scanner Evidence & Audit Trail

trivy (SCA)

FINDINGS

Scan Scope: Scanned 1 target(s): filesystem

Files Analyzed: 1

Rules Checked: 4

Checks Performed:

- Vulnerability scanning
- Misconfiguration detection
- Secret detection
- License compliance

Standards: CVE/NVD, OWASP Top 10 (A06), CIS Benchmarks, NIST SP 800-53

Methodology:

Scans filesystem for package manifests and lock files, matches installed packages against vulnerability databases (NVD, GitHub Advisory, vendor-specific). Also checks for misconfigurations and secrets using built-in rules.

Targets Analyzed:

Loose File License(s)

gitleaks (Secrets)

PASS

Scan Scope: Working directory scanned for 150+ secret patterns with entropy analysis

Configuration: Custom gitleaks.toml

Files Analyzed: 148

Rules Checked: 150

Checks Performed:

- AWS credential detection
- API key detection
- Private key detection
- Generic secret patterns
- High-entropy string analysis
- Token detection

Standards: CWE-798, CWE-259, OWASP Top 10 (A07)

Methodology:

Regex-based pattern matching against 150+ secret patterns with entropy analysis. Scans current file contents and optionally git commit history. Applies allowlists and custom rules to reduce false positives. Secrets are redacted in output.

opengrep (SAST)

FINDINGS

Scan Scope: Multi-language SAST with auto, security-audit, owasp-top-ten, cwe-top-25, r2c-security-audit, and secrets rulesets

Configuration: Custom semgrep-rules.yml + default rulesets

Files Analyzed: 27

Rules Checked: 2

Checks Performed:

- OWASP Top 10 patterns
- Security audit rules
- Secrets detection
- Injection patterns
- XSS patterns
- Insecure crypto usage

Standards: OWASP Top 10, CWE Top 25, SANS Top 25

Methodology:

Pattern-based static analysis using Semgrep rule engine with auto, security-audit, owasp-top-ten, and secrets rulesets. Matches AST patterns against known vulnerability signatures.

checkov (IaC)**PASS**

Scan Scope: IaC security analysis — 0 files, 0 checks (0 passed, 0 failed), frameworks: auto-detected

Configuration: Custom .checkov.yaml

Files Analyzed: 43

Checks Performed:

- Terraform misconfigurations
- CloudFormation template issues
- Kubernetes manifest hardening
- Dockerfile best practices
- Helm chart security
- GitHub Actions security
- GitLab CI security
- Secrets in IaC files

Standards: CIS Benchmarks, NIST SP 800-53, SOC 2, PCI-DSS, HIPAA

Methodology:

Graph-based static analysis of infrastructure-as-code files. Parses IaC definitions, builds resource graphs, and checks 1000+ built-in policies covering security, networking, encryption, and compliance.

grype (SCA)**PASS**

Scan Scope: SCA analysis: 0 unique CVEs across 0 vulnerable packages (filesystem)

Configuration: Grype with --by-cve deduplication and --add-cpes-if-none

Files Analyzed: 148

Checks Performed:

- CVE vulnerability matching
- Dependency version analysis
- CPE-based matching
- Advisory cross-referencing
- Fix availability detection

Standards: CVE/NVD, OWASP Top 10 (A06), GHSA

Methodology:

Generates SBOM using Syft, then matches all identified packages against vulnerability databases (NVD, GitHub Advisories, OS vendor databases). Reports CVEs with severity, fix versions, and references.

syft (SBOM)**FINDINGS**

Scan Scope: SBOM of 1 packages across 0 language(s): N/A

Configuration: Syft with JSON + CycloneDX output for SBOM standards compliance

Files Analyzed: 1

Rules Checked: 4

Checks Performed:

- SBOM generation (Syft JSON + CycloneDX)
- License compliance analysis

- Copyleft license detection (AGPL, GPL, MPL)
- Weak copyleft classification (LGPL)
- Unknown license flagging
- Package URL (PURL) generation
- CPE generation

Standards: NTIA SBOM Minimum Elements, CycloneDX, SPDX, Executive Order 14028

Methodology:

Catalogs software packages by parsing package manifests, lock files, and installed package databases. Generates standardized SBOM in CycloneDX or SPDX format for supply chain visibility.

package-validator (AI Code Quality)

N/A

Scan Scope: Dependency files (package.json, requirements.txt, go.mod) validated against package registries

Files Analyzed: 148

Checks Performed:

- npm registry existence validation
- PyPI registry existence validation
- Go module proxy existence validation
- Slopsquatting attack vector detection
- Hallucinated dependency identification

Standards: CWE-829, OWASP A08:2021, SLSA Supply Chain Security

Methodology:

Parses dependency manifests and validates each package exists in its respective registry (npm, PyPI, Go proxy) via HTTP HEAD requests. Non-existent packages are flagged as hallucinated — a common AI code generation artifact that creates supply chain attack vectors.

oxlint (Code Quality)

PASS

Scan Scope: JS/TS source files

Files Analyzed: 1

Checks Performed:

- JavaScript/TypeScript linting
- Code correctness checks
- Performance anti-patterns
- Suspicious code detection

Standards: CWE, ESLint Core Rules

Methodology:

Rust-based linter that performs AST analysis of JS/TS files at high speed. Detects correctness issues, suspicious patterns, and performance anti-patterns.

ruff (Code Quality)

FINDINGS

Scan Scope: Python linting and security analysis via Ruff

Configuration: Default Ruff rules

Files Analyzed: 2

Checks Performed:

- Pyflakes error detection
- pycodestyle enforcement
- flake8-bandit security checks
- isort import ordering
- pep8-naming conventions
- pyupgrade modernization
- flake8-bugbear bug detection
- flake8-comprehensions optimization

Standards: PEP 8, PEP 257, CWE, OWASP Top 10

Methodology:
Rust-based Python linter implementing 800+ rules from flake8, pylint, isort, and others. 10-100x faster than traditional Python linters.

actionlint (CI/CD Security)

PASS

Scan Scope: .github/workflows/*.yaml files

Files Analyzed: 43

Checks Performed:

- Workflow syntax validation
- Action version pinning checks
- Shell script validation
- Expression type checking
- Permissions analysis

Standards: GitHub Actions Security Best Practices, CWE-78

Methodology:
Static analysis of GitHub Actions workflow files. Validates syntax, checks for unpinned action versions, analyzes shell scripts for errors, and verifies expression types.

jscpd (Code Quality)

PASS

Scan Scope: All source files across supported languages

Files Analyzed: 3

Checks Performed:

- Cross-file duplicate detection
- Code clone identification
- Duplication percentage calculation

Standards: ISO/IEC 25010 (Maintainability)

Methodology:
Token-based analysis to detect copy-pasted code blocks across files and languages. Reports duplication percentage and identifies specific cloned regions.

typos (Code Quality)

FINDINGS

Scan Scope: All files in project directory

Files Analyzed: 148

Checks Performed:

- Source code spelling analysis
- Identifier typo detection
- Comment typo detection
- String literal typo detection

Standards: ISO/IEC 25010 (Maintainability)

Methodology:
Fast, low-false-positive spell checker designed for source code. Checks identifiers, strings, and comments against a curated dictionary.

Open Findings (20)

ruff

Code Quality | 2 findings | Ultra-fast Python linter (pyflakes, pycodestyle, flake8-bandit)

Severity	Title	File	CWE / Rule
MEDIUM	Ruff F401: `pathlib.Path` imported but unused	.../scripts/lib/audit_emitter.py:70	RUFF-F401

`pathlib.Path` imported but unused

Fix: Auto-fix available: Remove unused import: `pathlib.Path` (applicability: safe)

MEDIUM Ruff F401: `time` imported but unused
`time` imported but unused

.../scripts/lib/audit_emitter.py:66

RUFF-F401

Fix: Auto-fix available: Remove unused import: `time` (applicability: safe)

opengrep

SAST | 2 findings | Multi-language static analysis for security vulnerabilities

Severity	Title	File	CWE / Rule
MEDIUM	contains-bidirectional-characters This code contains bidirectional (bidi) characters. While this is useful for support of right-to-left languages such as Arabic or Hebrew, it can also be used to trick language parsers into executing code in a manner that is different from how it is displayed in code editing and review tools. If this is not what you were expecting, please review this code in an editor that can reveal hidden Unicode characters.	...s/lib/skill-mining-helpers.sh:538	CWE-94: Improper Contr

```
requires login
```

Fix: Review this finding and apply the appropriate fix based on the description: This code contains bidirectional (bidi) characters. While this is useful for support of right-to-left languages such as Arabic or Hebrew, it can also be used to trick language parsers into executing c

OWASP: A03:2021 - Injection | Rule: generic.unicode.security.bidi.contains-bidirectional-characters

MEDIUM dynamic-urllib-use-detected

.../scripts/lib/audit_emitter.py:362

CWE-939: Improper Auth

Detected a dynamic value being used with urllib. urllib supports 'file://' schemes, so a dynamic value controlled by a malicious actor may allow them to read arbitrary files. Audit uses of urllib calls to ensure user data cannot control the URLs, or consider using the 'requests' library instead.

```
requires login
```

Fix: Review this finding and apply the appropriate fix based on the description: Detected a dynamic value being used with urllib. urllib supports 'file://' schemes, so a dynamic value controlled by a malicious actor may allow them to read arbitrary files. Audit uses of urllib call

OWASP: A | Rule: python.lang.security.audit.dynamic-urllib-use-detected.dynamic-urllib-use-detected

syft

SBOM | 1 finding | Software bill of materials generator for supply chain visibility

Severity	Title	File	CWE / Rule
LOW	Unknown License: actions/checkout@11d5960a326750d5838... Package actions/checkout has no license information. This may indicate a compliance risk.	/.github/workflows/ci.yml	SBOM-LICENSE-UNKNOWN

```
Package: actions/checkout
Version: 11d5960a326750d5838078e36cf38b85af677262
Type: github-action
```

Fix: Verify the license for this package before using in production.

OWASP: A08:2021-Software and Data Integrity Failures

trivy

SCA | 1 finding | Container image and filesystem vulnerability scanner

Severity	Title	File	CWE / Rule
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LOW	License Compliance: Apache-2.0 in Package uses license Apache-2.0 (category: notice). Review for compliance with your organization's license policy.	LICENSE	LICENSE-Apache-2.0
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Package:
License: Apache-2.0
Category: notice

Fix: Review this finding and apply the appropriate fix based on the description: Package uses license Apache-2.0 (category: notice). Review for compliance with your organization's license policy.

OWASP: A06:2021-Vulnerable and Outdated Components

typos

Code Quality | 14 findings | Source code spell checker

Severity	Title	File	CWE / Rule
INFO	Typo: "BA" should be "BY, BE" Found typo "BA" in typo. Suggested corrections: BY, BE. Fix: Replace "BA" with "BY, BE".	agents/compliance.md:45	TYPOS-SPELLING
INFO	Typo: "RTO" should be "TO" Found typo "RTO" in typo. Suggested correction: TO. Fix: Replace "RTO" with "TO".	agents/research.md:491	TYPOS-SPELLING
INFO	Typo: "intoto" should be "into" Found typo "intoto" in typo. Suggested correction: into. Fix: Replace "intoto" with "into".	agents/supply-chain-security.md:106	TYPOS-SPELLING
INFO	Typo: "intoto" should be "into" Found typo "intoto" in typo. Suggested correction: into. Fix: Replace "intoto" with "into".	agents/supply-chain-security.md:46	TYPOS-SPELLING
INFO	Typo: "xdescribe" should be "describe" Found typo "xdescribe" in typo. Suggested correction: describe. Fix: Replace "xdescribe" with "describe".	agents/completeness-validator.md:690	TYPOS-SPELLING
INFO	Typo: "Hashi" should be "Hash" Found typo "Hashi" in typo. Suggested correction: Hash. Fix: Replace "Hashi" with "Hash".	agents/secrets-lifecycle.md:95	TYPOS-SPELLING
INFO	Typo: "Hashi" should be "Hash" Found typo "Hashi" in typo. Suggested correction: Hash. Fix: Replace "Hashi" with "Hash".	agents/secrets-lifecycle.md:29	TYPOS-SPELLING
INFO	Typo: "Hashi" should be "Hash" Found typo "Hashi" in typo. Suggested correction: Hash. Fix: Replace "Hashi" with "Hash".	agents/ciso.md:1381	TYPOS-SPELLING
INFO	Typo: "nformation" should be "information" Found typo "nformation" in typo. Suggested correction: information. Fix: Replace "nformation" with "information".	agents/ciso.md:662	TYPOS-SPELLING
INFO	Typo: "Hashi" should be "Hash" Found typo "Hashi" in typo. Suggested correction: Hash. Fix: Replace "Hashi" with "Hash".	templates/n8n-audit-emitter.json:103	TYPOS-SPELLING

INFO	Typo: "unparseable" should be "unparsable" Found typo "unparseable" in typo. Suggested correction: unparsable.	API.md:823	TYPOS-SPELLING
Fix:	Replace "unparseable" with "unparsable".		
INFO	Typo: "fpr" should be "for, far, fps" Found typo "fpr" in typo. Suggested corrections: for, far, fps.	...references/outcome-reports.md:46	TYPOS-SPELLING
Fix:	Replace "fpr" with "for, far, fps".		
INFO	Typo: "fpr" should be "for, far, fps" Found typo "fpr" in typo. Suggested corrections: for, far, fps.	...references/outcome-reports.md:13	TYPOS-SPELLING
Fix:	Replace "fpr" with "for, far, fps".		
INFO	Typo: "mis" should be "miss, mist" Found typo "mis" in typo. Suggested corrections: miss, mist.	docs/threat-model-v0.1.0.md:161	TYPOS-SPELLING
Fix:	Replace "mis" with "miss, mist".		

Codebase Coverage (148 files analyzed)

This scan analyzed 148 files using 11 security tools across 8 testing categories. Each scanner targets files relevant to its analysis domain (e.g., SAST scanners analyze source code, SCA scanners analyze dependency manifests).

Files by Language:

Markdown: 70 files	Config: 43 files
Shell: 24 files	Python: 2 files
JavaScript/TypeScript: 1 files	

Files by Extension:

.md: 69	.yaml: 26	.sh: 24
.json: 15	.other: 4	.py: 2
.gitignore: 2	.yml: 1	.js: 1
.tag: 1	.toml: 1	.txt: 1
.asc: 1		

Per-Scanner File Coverage:

Scanner	Category	Files Analyzed	Scope
trivy	SCA	1	Scanned 1 target(s): filesystem
gitleaks	Secrets	148	Working directory scanned for 150+ secret p...
opengrep	SAST	27	Multi-language SAST with auto, security-aud...
checkov	IaC	43	IaC security analysis — 0 files, 0 checks (...)
grype	SCA	148	SCA analysis: 0 unique CVEs across 0 vulner...
syft	SBOM	1	SBOM of 1 packages across 0 language(s): N/A
package-validator	AI Code Quality	148	Dependency files (package.json, requirement...
oxlint	Code Quality	1	JS/TS source files
ruff	Code Quality	2	Python linting and security analysis via Ruff
actionlint	CI/CD Security	43	.github/workflows/*.yml files
jscpd	Code Quality	3	All source files across supported languages
typos	Code Quality	148	All files in project directory

Files with Findings (15):

File Path	Findings	Scanners
agents/supply-chain-security.md	2	typos
agents/ciso.md	2	typos
hooks/scripts/lib/audit_emitter.py	2	ruff
skills/outcome-measurement/references/outcome-reports.md	2	typos
agents/secrets-lifecycle.md	2	typos
agents/completeness-validator.md	1	typos
agents/compliance.md	1	typos
agents/research.md	1	typos
docs/threat-model-v0.1.0.md	1	typos
/.github/workflows/ci.yml	1	syft
templates/n8n-audit-emitter.json	1	typos
/scan-target/hooks/scripts/lib/audit_emitter.py	1	opengrep
/scan-target/scripts/lib/skill-mining-helpers.sh	1	opengrep
API.md	1	typos
LICENSE	1	trivy

Software Bill of Materials (1 packages)

Package types: github-action: 1

Top licenses: Unknown (1)

Package	Version	Type	Language	License
actions/checkout	11d5960a326750	github-actio	-	Unknown